

AP CALCULUS

Stuff you MUST know Cold

* topic only on BC

Basic Derivatives

where u is a function of x , and a is a constant.

$$\frac{d}{dx}(x^n) = nx^{n-1}$$

$$\frac{d}{dx}(\sin x) = \cos x$$

$$\frac{d}{dx}(\cos x) = -\sin x$$

$$\frac{d}{dx}(\tan x) = \sec^2 x$$

$$\frac{d}{dx}(\cot x) = -\csc^2 x$$

$$\frac{d}{dx}(\sec x) = \sec x \tan x$$

$$\frac{d}{dx}(\csc x) = -\csc x \cot x$$

$$\frac{d}{dx}(\ln u) = \frac{1}{u} \frac{du}{dx}$$

$$\frac{d}{dx}(\log_a x) = \frac{1}{x \ln a}$$

$$\frac{d}{dx}(e^u) = e^u \frac{du}{dx}$$

$$\frac{d}{dx}(a^{u(x)}) = a^{u(x)} \ln a \cdot \frac{du}{dx}$$

$$\frac{d}{dx}(\sin^{-1} u) = \frac{1}{\sqrt{1-u^2}} \frac{du}{dx}$$

$$\frac{d}{dx}(\cos^{-1} x) = \frac{-1}{\sqrt{1-x^2}}$$

$$\frac{d}{dx}(\tan^{-1} u) = \frac{1}{1+u^2} \cdot \frac{du}{dx}$$

$$\frac{d}{dx}(\cot^{-1} x) = \frac{-1}{1+x^2}$$

$$\frac{d}{dx}(\sec^{-1} u) = \frac{1}{|u|\sqrt{u^2-1}} \cdot \frac{du}{dx}$$

$$\frac{d}{dx}(\csc^{-1} x) = \frac{-1}{|x|\sqrt{x^2-1}}$$

don't bother memorizing these!

Differentiation Rules

Chain Rule $\frac{d}{dx}[f(g(x))] = f'(g(x))g'(x)$

Product Rule $\frac{d}{dx}(uv) = u \frac{dv}{dx} + v \frac{du}{dx}$

Quotient Rule $\frac{d}{dx}\left(\frac{u}{v}\right) = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$

Some handy INTEGRALS:

$$\int \tan x \, dx = \ln|\sec x| + C$$

$$= -\ln|\cos x| + C$$

$$\int \sec x \, dx = \ln|\sec x + \tan x| + C$$

Curve sketching and analysis

$y = f(x)$ must be continuous at each:

critical point: $\frac{dy}{dx} = 0$ or undefined

inc/dec. function $f(x)$: $f' > 0, f' < 0$

concavity $\equiv$ inc/dec slope: $f'' > 0, f'' < 0$

local minimum:

$$\frac{dy}{dx} \text{ goes } (-, 0, +) \text{ or } (-, \text{DNE}, +) \text{ or } \frac{d^2y}{dx^2} > 0$$

local maximum:

$$\frac{dy}{dx} \text{ goes } (+, 0, -) \text{ or } (+, \text{DNE}, -) \text{ or } \frac{d^2y}{dx^2} < 0$$

point of inflection:

$f'' = 0$ or DNE AND concavity changes

$$\frac{d^2y}{dx^2} \text{ goes from } (+ \text{ to } -), (- \text{ to } +),$$

Abs. Max/Min: eval. crit # & endpts.

OR discuss "always inc or always dec."

Intermediate Value Theorem:

If the function $f(x)$ is continuous on $[a, b]$, for all k between $f(a)$ and $f(b)$, there exists at least one number $x = c$ in the open interval (a, b) such that $f(c) = k$.

Extreme Value Theorem:

If the function $f(x)$ is continuous on $[a, b]$, then there exists an absolute max and min on that interval.

Rolle's Theorem:

If the function $f(x)$ is continuous on $[a, b]$, AND differential on the interval (a, b) , AND $f(a) = f(b)$, then there is at least one number $x = c$ in (a, b) such that $f'(c) = 0$

Mean Value Theorem: $m_{\text{secant}} = m_{\text{tangent}}$

If the function $f(x)$ is continuous on $[a, b]$, AND differential on the interval (a, b) , then there is at least one number $x = c$ in (a, b) such that $f'(c) = \frac{f(b) - f(a)}{b - a}$.

MVT of Integrals i.e. AVERAGE VALUE:

If the function $f(x)$ is continuous on $[a, b]$ and differential on the interval (a, b) , then there exists at least one number $x = c$ on (a, b) such that

$$f(c)(b - a) = \int_a^b f(x) \, dx$$

Area rectangle = Net Area Integral

$$f(c) = \frac{1}{(b - a)} \int_a^b f(x) \, dx$$

This value $f(c)$ is the "average value" of the function on the interval $[a, b]$.

Limit Strategies: Factor and cancel,

Rationalize Numerator, u-sub, L'Hôpital's rules:

$$\lim_{x \rightarrow \infty} \frac{ax^2}{\sqrt{bx^2 + c}} = \frac{\sqrt{a}}{\sqrt{b}}$$

2 HA

$$\lim_{x \rightarrow \infty} \frac{ax}{\sqrt{bx^2 + c}} = \frac{a}{\sqrt{b}}$$

To find all HA: Take limit as $x \rightarrow \text{both } \pm \infty$

Approximation Methods for Integration

Use Geometry formulas

Rectangles - Left, Right and Middle

Riemann Sums $A = bh$

Trapezoids: $A = \frac{1}{2}(b_1 + b_2)h$

(Effects of inc/dec & concavity on approx.)

Concave up: M under estimate, T over estimate

Concave down: M over estimate, T under estimate

Inc: L=under, R=over. Dec: L=over, R=under

First Fundamental Th. of Calculus

$$\int_a^b f'(x) \, dx = f(b) - f(a)$$

2nd Fundamental Th. of Calculus

$$\frac{d}{dx} \int_a^{b(x)} f(t) \, dt =$$

$$f(b(x))b'(x) - f(a(x))a'(x)$$

Solids of Revolution and friends

Disk Method

$$V = \pi \int_a^b [R(x)]^2 \, dx$$

Washer Method

$$V = \pi \int_a^b ([R(x)]^2 - [r(x)]^2) \, dx$$

volume by cross section (not rotated)

$$V = \int_a^b \text{Area}(x) \, dx$$

$$A_{\text{eq. lat. } \Delta} = \frac{s^2}{4} \sqrt{3}$$

*Arc Length $s = \int_a^b \sqrt{1 + [f'(x)]^2} \, dx$

*Surface Area

$$SA = 2\pi \int_a^b \text{radius} \sqrt{1 + [f'(x)]^2} \, dx$$

Distance, Velocity, and Acceleration

velocity = $\frac{d}{dt}$ (position)

acceleration = $\frac{d}{dt}$ (velocity)

*velocity vector = $\left\langle \frac{dx}{dt}, \frac{dy}{dt} \right\rangle$

speed = $\|v\| = \sqrt{(x')^2 + (y')^2}$ *

displacement = $\int_{t_0}^t v \, dt$

total distance = $\int_{\text{initial time}}^{\text{final time}} |v| \, dt$

$$\int_a^b \sqrt{(x')^2 + (y')^2} \, dt *$$

Av. velocity = $\frac{1}{b-a} \int_a^b v(t) \, dt = \frac{s(b) - s(a)}{b-a}$

$$\frac{1}{b-a} \int_a^b \sqrt{[x'(t)]^2 + [y'(t)]^2} \, dt *$$

BC TOPICS and important TRIG identities and values:

L'Hôpital's Rule

$$\text{If } \frac{f(a)}{g(b)} = \frac{0}{0} \text{ or } \frac{\infty}{\infty},$$

$$\text{then } \lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$$

Euler's Method

If given that $\frac{dy}{dx} = f(x, y)$ and the solution passes through (x_o, y_o) ,

$$x_{\text{new}} = x_{\text{old}} + \Delta x$$

Basically Algebra I:

$$y = m(x - x_1) + y_1$$

$$y_{\text{new}} = \left(\frac{dy}{dx} \Big|_{(x_{\text{old}}, y_{\text{old}})} \right) (\Delta x) + y_{\text{old}}$$

Slope of a Parametric equation

Given $x(t)$ and $y(t)$, the slope is

$$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}}, \quad \frac{d^2y}{dx^2} = \frac{D_t \left(\frac{dy}{dt} \right)}{\frac{dx}{dt}}$$

Polar Curve

For a polar curve $r(\theta)$, the

$$\text{AREA inside a "leaf"} = \frac{1}{2} \int_{\theta_1}^{\theta_2} [r(\theta)]^2 d\theta$$

θ_1 and θ_2 are the "first" two times that $r = 0$.

$\frac{dr}{d\theta}$ determines inc/dec and relative max/mins.

The SLOPE of $r(\theta)$ at a given θ is

$$\frac{dy}{dx} = \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}} = \frac{\frac{d}{d\theta} [r(\theta) \sin \theta]}{\frac{d}{d\theta} [r(\theta) \cos \theta]}$$

Values of Trigonometric Functions for Common Angles

θ	$\sin \theta$	$\cos \theta$	$\tan \theta$
0°	0	1	0
$\frac{\pi}{6}$	$\frac{1}{2}$	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{3}}{3}$
$\frac{\pi}{4}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{2}}{2}$	1
$\frac{\pi}{3}$	$\frac{\sqrt{3}}{2}$	$\frac{1}{2}$	$\sqrt{3}$
$\frac{\pi}{2}$	1	0	" ∞ "
π	0	-1	0

Know both the **inverse trig** and the **trig** values.

Careful: $\tan\left(\frac{3\pi}{4}\right) = -1$ but $\arctan(-1) = -\frac{\pi}{4}$

Integration by Parts

($u=ILATE=du$)

$$\int u dv = uv - \int v du$$

$$\int f(x)g'(x)dx = f(x)g(x) - \int g(x)f'(x)dx$$

Integral of Ln

Use integration by parts and let $u = \ln x$

$$\int \ln x dx = x \ln x - x + C$$

nth Term: diverges if $\lim_{n \rightarrow \infty} a_n \neq 0$

Ratio Test: converges if $\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| < 1$

If the limit equals 1, you know nothing so check the endpoints using another test.

Root Test: converges if $\lim_{n \rightarrow \infty} \sqrt[n]{a_n} < 1$

p-series Test: $\sum_{n=1}^{\infty} \frac{1}{p^n}$ converges if $p > 1$

Alternating: converges if alt. & $\lim_{n \rightarrow \infty} a_n = 0$

Taylor Series

If the function f is "smooth" (continuous and differentiable) at $x = c$, then it can be approximated by the n^{th} degree polynomial

$$f(x) \approx f(c) + f'(c)(x-c) + \frac{f''(c)}{2!}(x-c)^2 + \dots + \frac{f^{(n)}(c)}{n!}(x-c)^n.$$

Geometric Series: $\sum_{n=0}^{\infty} a_1 r^n$ converges if $|r| < 1$

Limit Comparison: If $\lim_{n \rightarrow \infty} \frac{a_n}{b_n}$ exists and $\neq 0$,

what one series does (C or D), so does the other.

Direct Comparison: If $0 < a_n < b_n$ and

if $\sum_{n=1}^{\infty} a_n$ diverges, then $\sum_{n=1}^{\infty} b_n$ diverges;

if $\sum_{n=1}^{\infty} b_n$ converges then $\sum_{n=1}^{\infty} a_n$ converges.

Trig Identities

Double Angle

$$\sin 2x = 2 \sin x \cos x$$

$$\cos 2x = \cos^2 x - \sin^2 x = 1 - 2 \sin^2 x$$

Power Reduction

$$\sin^2 x = \frac{(1 - \cos 2x)}{2}$$

$$\cos^2 x = \frac{(1 + \cos 2x)}{2}$$

Pythagorean

$$\sin^2 x + \cos^2 x = 1$$

(others are easily derivable by dividing by $\sin^2 x$ or $\cos^2 x$)

$$1 + \tan^2 x = \sec^2 x$$

$$\cot^2 x + 1 = \csc^2 x$$

Reciprocal

$$\sec x = \frac{1}{\cos x} \text{ or } \cos x \sec x = 1$$

$$\csc x = \frac{1}{\sin x} \text{ or } \sin x \csc x = 1$$

Odd-Even

$$\sin(-x) = -\sin x \quad (\text{odd})$$

$$\cos(-x) = \cos x \quad (\text{even})$$

$$\tan(-x) = -\tan x \quad (\text{odd})$$

Maclaurin Series (Taylor Series about $x = 0$)

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots + \frac{x^n}{n!} + \dots$$

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots + \frac{x^{2n}(-1)^n}{(2n)!} + \dots$$

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots + \frac{x^{2n+1}(-1)^n}{(2n+1)!} + \dots$$

$$\frac{1}{1-x} = 1 + x + x^2 + x^3 + \dots + x^n + \dots$$

$$\ln(x) = \frac{(x-1)^1}{1} - \frac{(x-1)^2}{2} + \frac{(x-1)^3}{3} - \dots + \frac{(x-1)^n(-1)^{n+1}}{n} + \dots$$

Error Bound (Remainder)

Alternating Series:

If $S_N = \sum_{k=1}^N (-1)^k a_n$ is the N^{th} partial sum of a

convergent alternating series,

$$|R_N| \leq |a_{N+1}| \quad (\text{the next term})$$

Lagrange Error Bound of a Taylor Series:

Let $c = \#$ centered on and $x =$ value you want to approximate. There exists a z between x and c ,

$$\text{such that } |R_n| \leq \frac{f^{(n+1)}(z)}{(n+1)!} |x - c|^{n+1}$$

Find interval of $f^{(n+1)}(z)$ to find error interval.

Infinite Sums

Geometric Series: $S_{\infty} = \frac{a_1}{1-r}$ if $|r| < 1$

Telescoping Series: Expand & cancel

Special Series: $\sum_{n=0}^{\infty} \frac{1}{n!} = e^1$